

Python Logic Building

50 Practice Problems for Grades 9 & 10

Variables · Data Types · Operators · Conditionals · Loops

50 Problems	8 Sections	3 Difficulty Levels	Grade 9-10 Level
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Each problem includes a clear description, sample input/output, and a hint to guide your thinking. Try to solve every problem on your own before looking at the hint!

Difficulty Legend	● Easy — Directly applies one concept.	● Medium — Combines 2 concepts.	● Hard — Requires creative thinking.
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Section 1: Variables & Data Types

Q1 Sum of Two Numbers

Easy

Read two integers from the user and print their sum.

Sample Input	Sample Output
5 3	8

Hint: Use `int()` to convert input to integer.

#variables #int

Q2 Swap Two Variables

Easy

Read two numbers a and b, then print them after swapping their values.

Sample Input	Sample Output
10 20	a = 20 b = 10

Hint: Try swapping without a third variable using tuple unpacking.

#variables

Q3 Circle Area

Easy

Read the radius of a circle (float) and print its area rounded to 2 decimal places. Use $\pi = 3.14159$.

Sample Input	Sample Output
7	153.94

Hint: $\text{Area} = \pi * r * r$

#float #operators

Q4 Temperature Converter

Easy

Read a temperature in Celsius and convert it to Fahrenheit. Print the result rounded to 1 decimal place.

Sample Input	Sample Output
100	212.0

Hint: $F = (C * 9/5) + 32$

#float #operators

Q5 String Repetition

Easy

Read a string and a positive integer n. Print the string repeated n times.

Sample Input	Sample Output
ha 3	hahaha

Hint: Use the `*` operator on strings.

#string #operators

Section 2: Operators

Q6 Quotient and Remainder

Easy

Read two integers a and b. Print the quotient and remainder when a is divided by b.

Sample Input	Sample Output
17 5	Quotient: 3 Remainder: 2

Hint: Use // for integer division and % for modulo.

#operators

Q7 Even or Odd

Easy

Read an integer and print 'Even' if it is even, otherwise print 'Odd'.

Sample Input	Sample Output
14	Even

Hint: A number is even if $n \% 2 == 0$.

#operators #conditionals

Q8 Digit Extraction

Easy

Read a 3-digit number. Print its hundreds, tens, and units digits on separate lines.

Sample Input	Sample Output
374	Hundreds: 3 Tens: 7 Units: 4

Hint: Use // and % operators to extract digits.

#operators

Q9 Power Without **

Medium

Read base b and exponent e (positive integer). Calculate b raised to the power e WITHOUT using the ** operator. Print the result.

Sample Input	Sample Output
3 4	81

Hint: Multiply b by itself e times using a loop.

#operators #loops

Q1
0

Simple Interest

Easy

Read principal P, rate R (%), and time T (years). Print the Simple Interest rounded to 2 decimal places.

Sample Input	Sample Output
1000 5 3	150.00

*Hint: $SI = (P * R * T) / 100$*

#float #operators

Section 3: Conditional Statements

Q1
1

Grade Calculator

Easy

Read a score (0-100) and print the grade:

90-100 → A | 80-89 → B | 70-79 → C | 60-69 → D | below 60 → F

Sample Input	Sample Output
85	B

Hint: Use if-elif-else chain.

#conditionals

Q1
2

Largest of Three Numbers

Easy

Read three integers a, b, and c. Print the largest number among them.

Sample Input	Sample Output
12 45 30	45

Hint: Compare numbers using nested if or logical operators.

#conditionals

Q1
3

Leap Year Checker

Medium

Read a year. Print 'Leap Year' if it is a leap year, otherwise 'Not a Leap Year'.

A year is a leap year if divisible by 4, EXCEPT century years must be divisible by 400.

Sample Input	Sample Output
2000	Leap Year

Hint: Check divisibility by 400, 100, and 4 in the right order.

#conditionals #operators

**Q1
4****Triangle Validity****Medium**

Read three sides of a triangle. Print 'Valid Triangle' if they can form a triangle, otherwise 'Invalid Triangle'.

Sample Input	Sample Output
3 4 5	Valid Triangle

Hint: Sum of any two sides must be greater than the third side.

#conditionals #operators

**Q1
5****BMI Calculator****Easy**

Read weight (kg) and height (m) as floats. Calculate BMI and print the category:

BMI < 18.5 → Underweight | 18.5-24.9 → Normal | 25-29.9 → Overweight | >= 30 → Obese

Sample Input	Sample Output
70 1.75	Normal

Hint: BMI = weight / (height * height)

#float #conditionals

**Q1
6****Positive, Negative, or Zero****Easy**

Read an integer and print 'Positive', 'Negative', or 'Zero' accordingly.

Sample Input	Sample Output
-5	Negative

Hint: Use simple if-elif-else.

#conditionals

**Q1
7****Ticket Pricing****Medium**

A cinema charges:

Age < 5: Free | 5-12: Rs.150 | 13-64: Rs.300 | 65+: Rs.200

Read age and print the ticket price.

Sample Input	Sample Output
10	Rs.150

Hint: Use if-elif-else with age ranges.

#conditionals

Q1
8

Alphabet Case Checker

Easy

Read a single character. Print 'Uppercase', 'Lowercase', or 'Not an alphabet' accordingly.

Sample Input	Sample Output
G	Uppercase

Hint: Use .isupper(), .islower(), .isalpha() string methods.

#string #conditionals

Section 4: Nested Conditionals

Q1
9

Calculator

Easy

Read two numbers and an operator (+, -, *, /). Perform the operation and print the result. For division by zero print 'Error: Division by zero'.

Sample Input	Sample Output
10 4 /	2.5

Hint: Use nested if to handle division by zero inside the / branch.

#nested conditionals #operators

Q2
0

Login System

Medium

Correct username is 'admin' and password is 'python123'. Read username then password. Print 'Access Granted' if both are correct. Print 'Wrong Password' if only the password is wrong. Print 'User Not Found' if the username is wrong.

Sample Input	Sample Output
admin python123	Access Granted

Hint: Check username first in outer if, then check password in inner if.

#nested conditionals #string

Q2
1

Quadrant Finder

Medium

Read x and y coordinates (integers). Print which quadrant the point lies in, or 'On X-axis', 'On Y-axis', or 'Origin' if applicable.

Sample Input	Sample Output
-3 5	Quadrant II

Hint: Check for origin and axes first, then determine quadrant.

#nested conditionals

Q2
2

Electricity Bill

Medium

Read units consumed. Calculate the bill:

0-100 units: Rs.5/unit

101-300 units: Rs.5 for first 100 + Rs.8/unit for next

Above 300: previous + Rs.10/unit above 300

Print total bill.

Sample Input	Sample Output
250	Rs . 1700

Hint: Use nested or chained if-elif to calculate slab-wise billing.

#nested conditionals #operators

Section 5: For Loop

Q2
3

Multiplication Table

Easy

Read an integer n. Print its multiplication table from 1 to 10.

Sample Input	Sample Output
7	7 x 1 = 7 7 x 2 = 14 ... 7 x 10 = 70

Hint: Use for i in range(1, 11).

#for loop

Q2
4

Sum of N Numbers

Easy

Read n, then read n integers one by one. Print their sum.

Sample Input	Sample Output
4 10 20 30 40	100

Hint: Initialize sum = 0 before the loop.

#for loop

Q2
5**Factorial****Easy**

Read a non-negative integer n . Print $n!$ (factorial).

Sample Input	Sample Output
6	720

Hint: Initialize result = 1 and multiply from 1 to n .

#for loop

Q2
6**Count Vowels****Easy**

Read a string. Count and print the number of vowels (a, e, i, o, u — both cases).

Sample Input	Sample Output
Hello World	3

Hint: Iterate each character and check if it's in 'aeiouAEIOU'.

#for loop #string

Q2
7**Reverse a String****Easy**

Read a string and print it reversed.

Sample Input	Sample Output
Python	nohtyP

Hint: Loop from last index to 0, or build a new string.

#for loop #string

Q2
8**Sum of Digits****Easy**

Read a positive integer. Print the sum of its digits.

Sample Input	Sample Output
4567	22

Hint: Convert to string and iterate, or use % 10 in a loop.

#for loop #operators

Q2
9

FizzBuzz

Easy

Print numbers from 1 to n. For multiples of 3 print 'Fizz', for multiples of 5 print 'Buzz', for multiples of both print 'FizzBuzz', otherwise print the number. Read n.

Sample Input	Sample Output
15	1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz

Hint: Check divisibility by 15 first, then 3, then 5.

#for loop #conditionals

Q3
0

Number Guessing Count

Medium

Read the secret number s and n guesses one by one. Count how many guesses exactly equal s. Print the count.

Sample Input	Sample Output
7 5 7 3 7 9	2

Hint: Maintain a counter and increment when guess == s.

#for loop #conditionals

Section 6: While Loop

Q3
1**Countdown****Easy**

Read a positive integer n . Print a countdown from n to 1, then print 'Blast off!'.

Sample Input	Sample Output
5	5 4 3 2 1 Blast off!

Hint: Decrement n by 1 each iteration. Stop when $n < 1$.

#while loop

Q3
2**Collatz Sequence****Medium**

Read a positive integer n . Apply the Collatz rule repeatedly until $n = 1$:

If n is even $\rightarrow n = n // 2$

If n is odd $\rightarrow n = 3*n + 1$

Print each value including the starting number.

Sample Input	Sample Output
6	6 3 10 5 16 8 4 2 1

Hint: Keep looping while $n \neq 1$.

#while loop #conditionals

Q3
3**ATM Simulation****Medium**

Balance starts at Rs.5000. Read withdrawal amounts repeatedly until -1 is entered. Print 'Success' for valid withdrawals, 'Insufficient Funds' if balance is too low. Print final balance at the end.

Sample Input	Sample Output
1000 3000 2000 -1	Success Success Insufficient Funds Balance: Rs.1000

Hint: Use a while True loop and break on -1.

#while loop #conditionals

Q3
4

Guess the Number Game

Medium

The secret number is 42. Read guesses one by one. Print 'Too High', 'Too Low', or 'Correct! You got it in X tries.' Count the tries.

Sample Input	Sample Output
50 30 42	Too High Too Low Correct! You got it in 3 tries.

Hint: Keep a tries counter and loop until guess == 42.

#while loop #conditionals

Q3
5

GCD (Euclidean Algorithm)

Medium

Read two positive integers. Find and print their Greatest Common Divisor (GCD) using the Euclidean algorithm.

Sample Input	Sample Output
48 18	6

Hint: While $b \neq 0$: $a, b = b, a \% b$. Then GCD is a .

#while loop #operators

Q3
6

Digit Counter

Easy

Read a positive integer. Count and print the number of digits it contains using a while loop (do NOT convert to string).

Sample Input	Sample Output
38472	5

Hint: Divide by 10 and count until the number becomes 0.

#while loop #operators

Q3
7

Armstrong Number Checker

Medium

Read a 3-digit number. Print 'Armstrong' if it is an Armstrong number (sum of cubes of its digits equals the number), otherwise print 'Not Armstrong'.

Sample Input	Sample Output
153	Armstrong

Hint: $153 = 1^3 + 5^3 + 3^3 = 153$

#while loop #operators

Section 7: Nested Loops

Q3
8

Right-Angle Triangle

Easy

Read n. Print a right-angled triangle of stars with n rows.

Sample Input	Sample Output
5	<pre>* ** *** **** *****</pre>

Hint: Outer loop for rows, inner loop prints stars.

#nested loops

Q3
9

Number Pattern

Easy

Read n. Print a pattern where row i contains the number i repeated i times.

Sample Input	Sample Output
4	<pre>1 22 333 4444</pre>

Hint: Inner loop runs i times and prints i without newline.

#nested loops

Q4
0

Pyramid Pattern

Medium

Read n. Print a centered star pyramid of height n.

Sample Input	Sample Output
4	<pre> * *** ***** *****</pre>

Hint: Row i has (n-i) spaces and (2*i-1) stars.

#nested loops

Q4
1

Prime Numbers in Range

Medium

Read two integers a and b. Print all prime numbers between a and b (inclusive).

Sample Input	Sample Output
10 30	11 13 17 19 23 29

Hint: For each number, check divisibility from 2 up to its square root.

#nested loops #conditionals

Q4
2

Multiplication Table Grid

Medium

Read n. Print an n x n multiplication table grid.

Sample Input	Sample Output
3	1 2 3 2 4 6 3 6 9

Hint: Use nested loops; print each value with a tab or fixed width.

#nested loops

Q4
3

Hollow Rectangle

Medium

Read rows r and columns c. Print a hollow rectangle of stars (only border stars, inside is space).

Sample Input	Sample Output
4 6	***** * * * * *****

Hint: Print * for border positions (first/last row or first/last column), else print space.

#nested loops #conditionals

Q4
4

Sum of Each Row

Medium

Read n rows of integers. Each row contains exactly 4 numbers. Print the sum of each row.

Sample Input	Sample Output
3 1 2 3 4 5 6 7 8 9 10 11 12	Row 1: 10 Row 2: 26 Row 3: 42

Hint: Use split() to read numbers from each line.

#nested loops

Section 8: Mixed / Challenge Problems

Q4
5

Palindrome Checker

Medium

Read a string. Print 'Palindrome' if it reads the same forwards and backwards, otherwise print 'Not Palindrome'. Ignore case.

Sample Input	Sample Output
Racecar	Palindrome

Hint: Convert to lowercase, then compare with its reverse.

#string #loops #conditionals

Q4
6

Caesar Cipher

Hard

Read a message (lowercase letters only) and a shift k. Encode the message by shifting each letter forward by k positions in the alphabet (wrapping around after z). Print the encoded message.

Sample Input	Sample Output
hello 3	khoor

Hint: Use `ord()` and `chr()`. $\text{New char} = \text{chr}((\text{ord}(c) - \text{ord}('a') + k) \% 26 + \text{ord}('a'))$

#string #loops #operators

Q4
7

Number to Binary

Medium

Read a positive integer. Print its binary representation WITHOUT using `bin()` or any built-in conversion.

Sample Input	Sample Output
13	1101

Hint: Repeatedly divide by 2 and collect remainders. Reverse at the end.

#while loop #operators #string

Q4
8

Star Diamond

Hard

Read an odd integer n. Print a diamond shape of stars.

Sample Input	Sample Output
5	<pre>* *** ***** *** *</pre>

Hint: Print upper half (including middle) then lower half separately using nested loops.

#nested loops #conditionals

Q4
9

Count Words

Medium

Read a sentence. Count and print the number of words in it.

Sample Input	Sample Output
I love Python programming	4

Hint: Split the string on spaces and count the parts.

#string #loops

Q5
0

Perfect Number Checker

Hard

Read a positive integer n. Print 'Perfect' if n equals the sum of all its proper divisors (divisors excluding n itself), otherwise print 'Not Perfect'.

Example: $28 = 1 + 2 + 4 + 7 + 14$

Sample Input	Sample Output
28	Perfect

Hint: Loop from 1 to n-1 and collect divisors (i where $n \% i == 0$).

#for loop #operators #conditionals

Practice makes perfect! Work through these problems step-by-step. Start with paper and pencil before coding.